

Question

Using triphenylmethyl tetrakis(pentafluorophenyl)borate ($\text{Ph}_3\text{CB}(\text{C}_6\text{F}_5)_4$) as an initiator, polyvinyl chloride ($-\text{[CH}_2\text{CHCl]}_n-$) can react with triethylsilane in chlorobenzene, where most of the chlorine on the polyvinyl chloride is reduced to hydrogen by triethylsilane, and the remaining chlorine is replaced by chlorobenzene, which acts as a solvent, to form $-\text{C}_6\text{H}_4\text{Cl}$, eventually generating a polymer product similar to ethylene-chlorostyrene copolymer.

The product was dissolved in deuterated tetrachloroethane for high-temperature NMR hydrogen spectroscopy. The spectrum only shows signals in the following chemical shift (δ) ranges:

- $\delta = 0$, single peak, peak area is 0.24
- $\delta = 0.5 \sim 3.5$, multiple peaks, peak area is 10.88
- $\delta = 6.0$, single peak, peak area is 0.96
- $\delta = 6.8 \sim 7.5$, multiple peaks, peak area is 1.00

Which of the following statements is correct:

- A.** All other options are incorrect
- B.** The reaction follows a radical mechanism
- C.** The proportion of $-\text{[CH}_2\text{CH}(\text{C}_6\text{H}_4\text{Cl})]-$ units in the product is 8.98%
- D.** The proportion of $-\text{[CH}_2\text{CH}(\text{C}_6\text{H}_4\text{Cl})]-$ units in the product is 8.42%
- E.** If anisole is used as the solvent, a similar reaction can also occur.
- F.** The proportion of $-\text{[CH}_2\text{CH}_2]-$ units in the product is 82.8%.

Answer

Correct Answer: **C**

Detailed Explanation

The Ph_3C^+ in $\text{Ph}_3\text{CB}(\text{C}_6\text{F}_5)_4$ can abstract H from triethylsilane to generate Et_3Si^+ , which as a Lewis acid combines with the chlorine of $-\text{[CH}_2\text{CHCl]}_n-$ on polyvinyl chloride to generate a carbocation. If anisole is used as a solvent, the O in it will interfere with the formation of silyl cation and carbocation intermediates, preventing the reaction from proceeding normally. Therefore, options B and E are incorrect.

CHECKPOINT

1 PTS

The reaction is a carbocation process

CHECKPOINT

1 PTS

When anisole is used as a solvent, the O in it will interfere with the formation of silyl cation and carbocation intermediates

The carbocation can be reduced to $-\text{H}$ by HSiEt_3 , or it can undergo a Friedel-Crafts reaction with chlorobenzene as a solvent, eventually forming a polymer product similar to an ethylene-chlorostyrene copolymer. The product can be represented as:



where x is the number of $-\text{[CH}_2\text{CH}_2\text{]}-$ units, and y is the number of $-\text{[CH}_2\text{CH(C}_6\text{H}_4\text{Cl)]}-$ units.

CHECKPOINT

1 PTS

The chemical formula of the product can be represented as $-\text{[CH}_2\text{CH}_2\text{]}_x-\text{[CH}_2\text{CH(C}_6\text{H}_4\text{Cl)]}_y-$

The high-temperature nuclear magnetic resonance hydrogen spectrum of the product was tested using deuterated tetrachloroethane as a solvent. The spectrum has signals in four chemical shift (δ) ranges, where:

- The peak around 0.0 comes from the internal standard substance.
- The peak around 6.0 is the solvent peak of deuterated tetrachloroethane.
- The peak at 0.5 ~ 3.5 comes from H connected to the alkyl group. There are 4 such H in each ethylene unit and 3 in each chlorostyrene unit.
- The peak at 6.8 ~ 7.5 comes from H connected to the benzene ring. There are 4 such H in each chlorostyrene unit.

CHECKPOINT

1 PTS

The peak around 0.0 comes from the internal standard substance

CHECKPOINT

1 PTS

The peak around 6.0 is the solvent peak of deuterated tetrachloroethane

CHECKPOINT

1 PTS

Only the signals at $0.5 \sim 3.5$ and $6.8 \sim 7.5$ in the nuclear magnetic resonance hydrogen spectrum come from the product

From the peak area of the nuclear magnetic resonance hydrogen spectrum, we know:

$$4x + 3y = kS_{0.5 \sim 3.5}$$

$$4y = kS_{6.8 \sim 7.5}$$

Solving the equations:

$$\frac{y}{x + y} = \frac{4S_{6.8 \sim 7.5}}{4S_{0.5 \sim 3.5} + S_{6.8 \sim 7.5}} = \frac{4 \times 1.00}{4 \times 10.88 + 1.00} = 0.0898$$

Therefore, the proportion of $-\text{CH}_2\text{CH}(\text{C}_6\text{H}_4\text{Cl})-$ units in the product is 8.98%, and the proportion of $-\text{CH}_2\text{CH}_2-$ units is 91.0%.

CHECKPOINT

1 PTS

The proportion of $-\text{CH}_2\text{CH}(\text{C}_6\text{H}_4\text{Cl})-$ units in the product is approximately 8.98%

CHECKPOINT

1 PTS

The proportion of $-\text{[CH}_2\text{CH}_2]-$ units in the product is 91.0%